

Section 5.1 Riemann Sums

$$\Sigma = \text{sum}$$

$$\overset{\text{stop} \rightarrow 5}{\sum_{i=1}^{\quad}} a_i = a_1 + a_2 + a_3 + a_4 + a_5$$

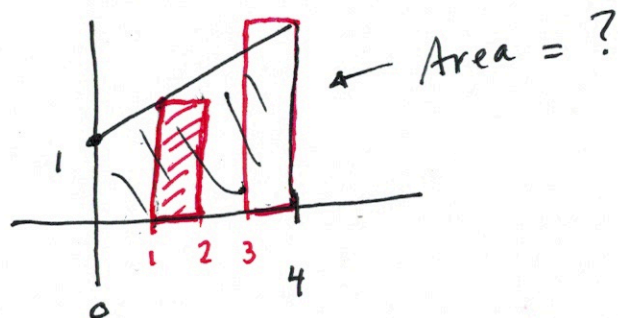
$\uparrow$
start

$$\text{Ex } \textcircled{1} \quad \sum_{n=1}^4 n^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30 \checkmark$$

$$\text{sum}(\text{seq}(x^2, x, 1, 4, 1))$$

$\uparrow$ function $\uparrow$ start $\uparrow$ step

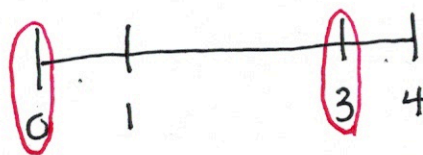
Ex ② $f(x) = 2x + 1$



Estimate area using 4 rectangles.

A) left - endpoints

$$\Delta x = \frac{4-0}{4} = 1$$



$$\text{Sum}(\text{seg}((2x+1) \cdot 1, x, 0, 3, 1)) = \textcircled{16}$$

$f(x) \cdot \Delta x, x, \text{start}, \text{stop}, \Delta x$

Ex (2) B) right - endpoints

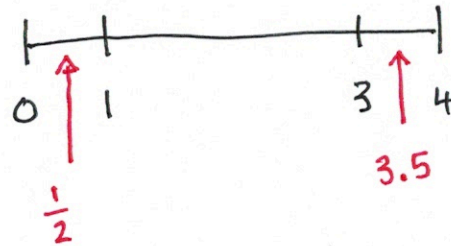
$$\Delta x = 1$$



$$\text{sum}(\text{seg}((2x+1) \cdot 1, x, 1, 4, 1)) = \boxed{24}$$

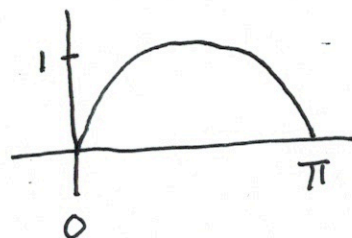
C) mid points

$$\Delta x = 1$$



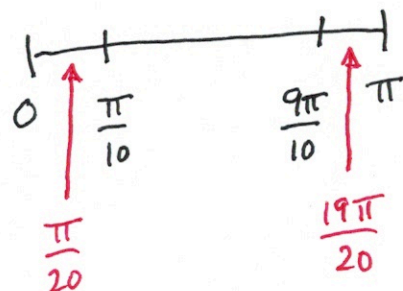
$$\text{sum}(\text{seg}((2x+1) \cdot 1, x, 0.5, 3.5, 1)) = \boxed{20}$$

Ex (3) $y = \sin x$ on $[0, \pi]$



Use midpoints of 10 rectangles.

$$\Delta x = \frac{\pi - 0}{10} = \frac{\pi}{10}$$



$$\text{Sum} \left(\text{seg} \left(\sin x \cdot \frac{\pi}{10}, x, \frac{\pi}{20}, \frac{19\pi}{20}, \frac{\pi}{10} \right) \right)$$

$$\approx 1.959$$

$$\text{Actual area} = 2$$